

CS 578: CYBER-SECURITY

INTRODUCTION

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Oregon State
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TRUE AI
Trustworthy and Responsible AI

SECURITY MINDSET

THE ANT FARM – FROM SCHNEIER’S

Uncle Milton Industries has been selling ant farms to children since 1956. Some years ago, I remember opening one up with a friend. There were no actual ants included in the box. Instead, there was a card that you filled in with your address, and the company would mail you some ants.

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- How do people look at the world?

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- How do people look at the world?
- How do security people look at the world?

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- One person saw an awesome feature
- But the other saw an unauthenticated request with no trust boundary

THE SECURITY MINDSET

- Most practitioners ask: how to make it work?
- Most security people ask: how do I make it **NOT** work?

THE SECURITY MINDSET – CONT'D

- Most practitioners ask: how to make it work?
- Most security people ask: how do I make it **NOT** work?
- Can it be learned?
 - Schneier isn't sure: how much is innate; how much is “teachable”?
 - One question open: this 10-week course bets on “teachable”

BUT THE MINDSET IS NOT ENOUGH – ANDERSON'S

- Many systems are built much more secure
 - Memory safety languages like Rust
 - Many secure authentication protocols
 - Least privilege principles in systems programming
 - Formal verifications
 - ... (many more)
- But
 - *Why doesn't any of it get deployed all the time?*

ANDERSON'S – INCENTIVES ARE IMPORTANT

- This is about bad incentives, not bad engineering
- His observations
 - Externalities: the fixer is not the payer (ATM fraud)

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 - Network effects: the first is important than being safe
 - Free riding: DDoS botnets
- It's about a discrepancy
 - Who *could* fix it?
 - Who actually *pays* for the fix?

MOVING FORWARD

- Think about two additional questions
 - Schneier: Is the attack practical? Where is the trust boundary?
 - Anderson: who would actually deploy the defense? and why would they bother?

HERE IS HOW YOU'LL LEARN

ABOUT ME



Who am I?

- Assistant Professor of Computer Science at OSU (Sep. 2021 ~)
- Ph.D. from the University of Maryland, College Park
- B.S. from Seoul National University, South Korea

What I do?

- **Formal:** I work at the intersection of security, privacy, and machine learning
- **Informal:** I am “AI-hacker”

What do I teach?

- Grad: CS499/579: Trustworthy ML | CS578: Cyber-security
- UGrad: CS374: Operating Systems I | CS370 : Introduction to Security

Where can you find me?

- **Email:** sanghyun.hong (at) oregonstate.edu | **Office:** 2029 KEC

MINDSETS NEEDED FOR THIS CLASS

- You are graduate students
 - Self-discipline (or in other words, independence)
 - Intellectual curiosity (or in other words, motivation to study)
 - (Pro)active learning
 - Respect

OVERVIEW

- Course overview:
 - 4 credit courses: 12 hours of effort per week
 - Course website: <https://true-lab.ai/courses/Sec-Grad/current/index.html>
 - All submissions: Canvas
- Contacts:
 - Instructor
 - Email to sanghyun.hong@oregonstate.edu
 - If you have personal matters or any questions/concerns related to the course
 - Office hours: Mondays 11 am – 12 pm (on Zoom, right before class)
 - No TA

OVERVIEW

- Computing resources:
 - OSU HPC: <https://it.engineering.oregonstate.edu/hpc>
 - OSU EECS: <https://eecs.oregonstate.edu/eecs-it#Servers>
 - **[Note]**
 - Read the OSU HPC websites carefully and request access to these resources
 - GPU computing resources can be crowded. If your project needs GPUs, start early

LEARNING OBJECTIVES

- You'll learn in this class
 - **[Security]** Security mindset: learn how to think like an adversary
 - **[Research]**
 - To learn emerging attacks and defenses
 - To learn how to pursue a research problem
 - To learn how to communicate your research findings with others
 - **[Practice]**
 - To have hands-on experience by implementing security mechanisms
- After taking this class, you'll
 - Be able to start research on cyber-security
 - Be ready for offering a security (or privacy) angle to the world!

COURSE STRUCTURE

- 10-week schedule; no textbook
 - Course syllabus is up: <https://true-lab.ai/courses/Sec-Grad/current/syllabus.html>
 - **Part I:** Network/Internet Security
 - **Part II:** Computer Systems Security
 - **Part III:** Isolation and Breaks
 - **Part IV:** Software/Web Security
 - **Part V:** Trustworthy ML

Schedule

[Note] This is a tentative schedule; subject to change depending on the progress.

Date	Topics	Notice(s)	Readings
Overview of Security Principles			
Wed. 09/23	Introduction [Slides]	Online	(Classic) The Security Mindset (Classic) Why Information Security is Hard – An Economic Perspective
Part I: Network/Internet Security			
Mon. 09/28	Internet Protocols [Slides]	In-person	(Classic) Censys: A Search Engine Backed by Internet-Wide Scanning (Classic) Mining Your Ps and Qs: Detection of Widespread Weak Keys in Network Devices
Wed. 09/30	Ecosystems [Slides]	Online [Team-up! by 09/30]	(Classic) A Longitudinal, End-to-End View of the DNSSEC Ecosystem (Classic) Analysis of SSL Certificate Reissues and Revocations in the Wake of Heartbleed
Mon. 10/05	Applications [Slides]	In-person	(Classic) The Science of Guessing: Analyzing an Anonymized Corpus of 70 Million Passwords (Recent) Analyzing the End-to-end Life Cycle and Effectiveness of Phishing ...
Wed. 10/07	-	[No class]	Checkpoint I Presentation Prep.

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 - **Part I:** Network/Internet Security
 - **Part II:** Computer Systems Security
 - **Part III:** Isolation and Breaks
 - **Part IV:** Software/Web Security
 - **Part V:** Trustworthy ML
- Hybrid course structure
 - Mon classes = In-person , OWEN 106
 - Wed classes = Online, Zoom (including today)
 - Term-project presentations = Must be in-person, 10/12, 11/09, and 12/02
 - Others: make sure to check the syllabus or the Canvas announcements

COURSE STRUCTURE – CONT'D

- In this course, you will do
 - 20%: written paper critiques
 - 15%: In-class paper presentation
 - Must complete sign-ups by 09/30
 - Sign-up form [link](#) on course website/Canvas
 - 50%: Term-project
 - Must complete team-ups by 09/30
 - 15%: Final Exam (take-home, on Canvas, 12/07)
- [**~10%**] extra credit opportunities
 - +5%: Outstanding project work
 - +5%: Submitting the final report to workshops

20%: WRITTEN PAPER CRITIQUES

- **[Due]** Before each class (hard deadline)
- You need to:
 - Pick a paper
 - Submit your review on **Canvas**

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20%: WRITTEN PAPER CRITIQUES

- **[Due]** Before each class (hard deadline)
- You need to:
 - Pick a paper
 - Submit your review on **Canvas**
 - Your review should include ([example](#))
 - Summary
 - Contributions
 - Weaknesses
 - Strengths
 - Your opinions

15%: IN-CLASS PAPER PRESENTATION

- You need to *sign-in* for this opportunity
 - Sign-up deadline: 09/30
 - First-come, first-served; No slot left = 0 points
 - Up to 2 students can present the same paper together
 - Use Google sheet to sign-up (link is available on Canvas and on the website)
- You **MUST** meet me **Once**:
 - ~0.5 weeks before the class for organizing your presentation
 - Missing this meeting will result in -5% points
- Structure
 - 25 min. paper presentation
 - 10-25 min. in-depth discussion

20%: WRITTEN PAPER CRITIQUES

- **[Due]** Before each class (hard deadline)
- You need to:
 - Pick a paper
 - Submit your review on **Canvas**
 - Your review should include ([example](#))
 - Summary, contributions, weaknesses, strengths, your opinions
 - **[Note]**
 - This is **not** a pleasant reading
 - Do **not** use AI to generate any content
 - Grades: 0 / 1 / 2

15%: IN-CLASS PAPER PRESENTATION – CONT'D

- Grading policy
 - 0%: No sign-up or no presentation
 - 8%: Sign-up; but the lack of quality in your presentation
 - 12%: Sign-up; you present a paper and lead the discussion
 - 15%: You make an outstanding presentation and discussion.
 - [Note] Must meet with me once before:
 - ~0.5 weeks before the class for organizing your presentation
 - Missing this meeting will result in -5% points

50%: TERM PROJECT

- You will form a team up to 3 students
 - Solo project are **NOT** allowed; must form a team
 - Use Google sheet to sign-up
 - The same link used to sign-up class presentations
 - **Deadline 09/30, -2%** if you are not on the sheet
- Project Topics
 - **Track 1:** Choose your own topic
 - **Track 2:** Implement a chosen SoTA LLM distillation prevention and tracing method
- Presentations
 - Checkpoint Presentation 1 (10/12, 15%)
 - Checkpoint Presentation 2 (11/09, 15%)
 - Final Presentation and a write-up (12/02, 20%)

COURSE STRUCTURE — RECAP

- In this course, you will do
 - 20%: written paper critiques
 - 15%: In-class paper presentation
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 - Sign-up form [link](#) on course website/Canvas
 - 50%: Term-project
 - Must complete team-ups by 09/30
 - 15%: Final Exam (take-home, on Canvas, 12/07)
- [**~10%**] extra credit opportunities
 - +5%: Outstanding project work
 - +5%: Submitting the final report to workshops

“GENEROUS” GRADING POLICY

- A : $\geq 90\%$
- A : $\geq 87\%$
- B+: $\geq 85\%$
- B : $\geq 80\%$
- B- : $\geq 77\%$
- C+: $\geq 75\%$
- C : $\geq 70\%$
- C- : $\geq 67\%$
- D+: $\geq 65\%$
- D : $\geq 60\%$; Below this threshold will lead to failure

LATE SUBMISSION POLICY

- Written paper critiques:
 - No submission in any case: 0 pts
- Term Project
 - No presentation in any cases: 0 pts
 - No report submission: 0 pts
- Final Exam:
 - No submission in any case: 0 pts

KEEP AN EYE ON THE COURSE WEBSITE AND CANVAS

- You will find updates such as:
 - New announcements
 - Changes in our course schedule (or structure)

Thank You!

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